

Required communication performance for unmanned aerial vehicles in non-terrestrial networks

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Abstract—Safe integration of drones in non-segregated is one of the key objectives for 2030. Moreover, airspace presents demanding communication requirements in the Command and Control (C2) link. This combines requirements from airspace and communication bodies. This work presents a systematic methodology to generate a Required Communication Performance (RCP), reinterpreting the current aviation methodology to this drone's air segment needs. We map safety requirements from aviation industry to key performance indicators of 3GPP networks. We explore how the current drones C2 link connected to terrestrial networks would benefit from the space segment. **We found that...**

Index Terms—UAV, RCP

I. INTRODUCTION

The safe integration of drones into non-segregated airspace needs a rigorous framework for defining link performance. One of the main objectives of the work is to develop a systematic approach to assessing command and control (C2) link reliability based on the mission. To achieve this, it is mandatory in aeronautical approaches to specify a Required Communication Performance (RCP) for an intended mission. The RCP specification for a drone is not a static value but a safety-derived requirement based on the most stringent communication transaction necessary for the mission's risk profile. This methodology adopts the Performance-Based Communication and Surveillance (PBCS) framework defined by ICAO [1] and adapted for Remotely Piloted Aircraft Systems (RPAS) by the Joint Authorities for Rulemaking on Unmanned Systems (JARUS) [2]–[4]. The rest of the work proceeds as follows: in Section II we develop the procedure to assess a RCP objective for the mission, in Section III we present the framework to validate the RCP both in terrestrial and non-terrestrial networks. Finally, in Section V we validate the RCP methodology by discussing the findings of both approaches.

II. METHODOLOGY

The proposed methodology follows a systematic four-phase approach: Mission Profiling, Transaction Analysis, Metric Specification, and System Allocation (summarized in Figure 1).

A. Mission and risk profiling

Even though our main goal is to define the performance metrics, first, we need to characterize the operational environment. Different operational environments will imply different levels of safety, thus different RCP specifications.

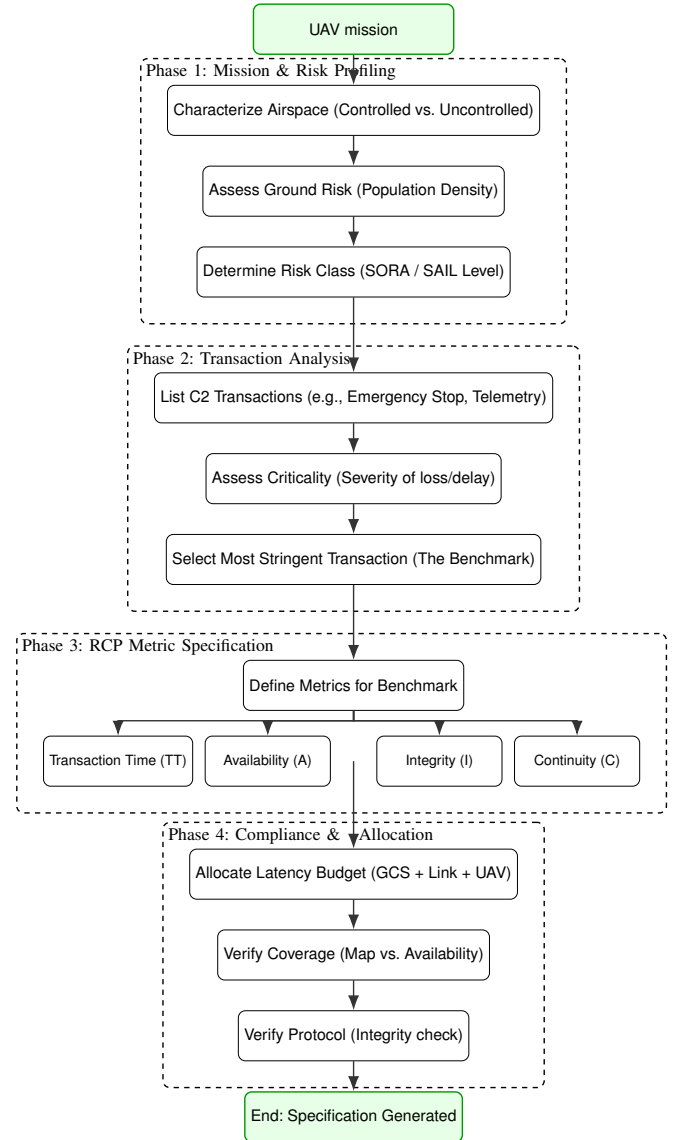


Fig. 1. Methodology for assessing an RCP

1) Airspace Characterization: The operating environment is classified to determine baseline separation standards. Missions occurring in Controlled Airspace (Classes A to C) impose stricter availability requirements due to the need for interaction with Air Traffic Management (ATM). Operations

in U-Space or uncontrolled airspace may rely on digital separation services.

2) *Risk Class Determination*: A Specific Assurance and Integrity Level (SAIL) is assigned to the mission based on two primary factors: the ground risk; quantified by population density (e.g., flight over sparse rural areas vs. urban gatherings), and the air risk, quantified by the density of airspace users and the likelihood of mid-air collision. As noted in [2], higher risk operations (e.g., Urban Beyond Visual Line of Sight - BVLOS) necessitate more stringent RCP types (e.g., RCP 10) to ensure a Time to Intervention (TTI) sufficient for safety. For example, a beyond-visual-line-of-sight drone flight over an urban area has a Ground Risk Class (GRC) above 5, regardless of the size of the drone, which results in a SAIL between IV and VI without mitigation measures. In OSO#6, this scenario is considered medium-to-high [5].

Since the SAIL level is medium-high, we already know we might need a stringent RCP. Prior to specify our objective metrics, we need to perform a transaction analysis.

B. Transaction Analysis

The fundamental unit of analysis in the JARUS C2 concept is the “Operational Communication Transaction”. [2]. An RCP specification is driven strictly by the most time-critical transaction required to maintain safety boundaries.

1) *Transaction Identification*: All critical Command and Control (C2) message exchanges must be cataloged. Common transactions include *Uplink* transactions (Emergency Motor Cut-off, Waypoint Update, Return-to-Home Command), and *Downlink* (System Heartbeat, Telemetry Update, Alert Propagation).

2) *Criticality and Benchmarking*: Each transaction is evaluated for the severity of its loss or delay. The transaction with the strictest latency requirement becomes the “Design Driver.” For instance, if a geofence breach requires a pilot intervention within 10 seconds to prevent an excursion from the operational volume, the entire C2 link is dimensioned to satisfy RCP 10, regardless of the laxer requirements of routine telemetry.

C. The RCP Metric Specification

The performance requirement is formally defined by four metrics: Availability (A), Transaction Time (TT), Continuity (C), and Integrity (I).

1) *Availability (A)*: Availability is the probability that the communication service is operational and capable of initiating a transaction at any random instant. For aviation safety, this ensures the link is present when an emergency command needs to be sent.

2) *Transaction Time (TT)*: The Transaction Time represents the maximum allowable time for the completion of a transaction with a specified confidence. It corresponds to the Expiration Time (ET) defined in ICAO PBCS, after which the initiator is required to revert to an alternative procedure [6].

$$TT = T_{HMI} + T_{Uplink} + T_{Proc} + T_{Downlink} \quad (1)$$

where T_{HMI} is the human-machine interface processing time, and T_{Proc} is the processing time on the UAS.

3) *Continuity (C)*: Continuity is the probability that a transaction will be completed within the Transaction Time (TT), given that the system was available at the start of the transaction. Klügel and Schupke [6] model this using the cumulative distribution function (CDF) of the transaction delay T :

$$C(TT) = \Pr\{T \leq TT\} = (1 - p_e) \cdot F_d(TT) \quad (2)$$

where p_e is the probability of packet error/loss, and $F_d(t)$ is the delay distribution function of the link. Target values typically range from 0.999 to 0.9999 per flight hour, depending on the SAIL.

4) *Integrity (I)*: Integrity is the probability of an undetected error occurring in a completed transaction (e.g., a UAV executes a “Turn Left” command when “Turn Right” was sent). This is strictly a function of the protocol’s error detection capabilities (e.g., CRC-32, cryptographic signatures) and is typically required to be $< 10^{-5}$ per flight hour [2].

D. Compliance and allocation

The final phase involves budgeting the derived requirements across the system architecture to ensure compliance.

1) *Latency Budget Allocation*: The total TT budget is shared to the subsystems. A typical allocation model is:

$$T_{CSP} \leq TT - (T_{GCS} + T_{Avionics} + T_{Pilot}) \quad (3)$$

This establishes the maximum latency Service Level Agreement (SLA) for the Communication Service Provider (CSP).

2) *Stochastic Verification*: Simple average latency is insufficient for safety-critical C2 links. As proposed in [6], the “tail” of the latency distribution must be modeled to verify that the probability of exceeding the TT satisfies the strict Continuity requirement $(1 - C(TT))$.

III. SYSTEM MODEL

Shreeja simulator

IV. USE CASE: RCP SPECIFICATION FOR URBAN BVLOS MEDICAL DELIVERY

To be adapted with Shreeja. We can keep the objectives if we both agree

This section demonstrates the application of the developed RCP methodology to a Beyond Visual Line of Sight (BVLOS) medical delivery mission, explicitly incorporating the Key Performance Indicators (KPIs) from 3GPP TS 22.125 Table 7.2-1.

A. Phase 1: Mission and Risk Profiling

1) Operational Concept (ConOps):

- **Mission**: Urgent transport of blood samples between two hospitals
- **Airspace**: Very Low Level (VLL) in a populated urban environment (U-Space).
- **Connectivity**: Two types: the first is a public 5G/LTE Network with dedicated QoS slicing. The second consist of an additional non-terrestrial segment (LEO satellite constellation).

2) Risk Assessment and Mode Selection:

- Target Safety Level: **SAIL IV**.
- 3GPP Control Mode: Steer to Waypoints (Automatic flight) as defined in 3GPP TS 22.125.

B. Phase 2: Transaction Analysis and 3GPP KPI Mapping

The mission requirements are mapped to the 3GPP standard functions from Table 7.2-1. The comparison of mission transactions versus 3GPP KPIs is detailed in Table I.

TABLE I
3GPP TS 22.125 KPI MAPPING

Transaction	3GPP Function	Latency	Reliability
Telemetry Update	UAV originated C2	500 ms	99.9%
Waypoint Update	UAV terminated C2	500 ms	99.9%
Manual Override	Direct stick steering	40 ms	99.9%
UTM Interaction	Automatic flight on UTM	5.0 s	99.9%

C. Phase 3: Merged Metric Specification

By synthesizing the safety-driven RCP methodology with the 3GPP network capabilities, we define the following profile for the BVLOS mission:

1) *Transaction Time (TT)*: According to Table s set to **500 ms** per 3GPP Steer to Waypoints specifications. However, the **RCP Safety Limit** (Expiration Time) remains **8.0 s**. If the 3GPP network fails to deliver within 8.0 s, the safety continuity is breached, triggering a fail-safe.

2) *Continuity and Reliability*: The 3GPP **99.9% Reliability** requirement is adopted as the **Continuity (C)** target. This specifies that 99.9% of messages must arrive within the 500 ms target window.

3) *Availability (A)*: While 3GPP focus is on per-message reliability, the SAIL IV mission requires **99.99% Availability**. This is achieved by combining the 3GPP network with redundant carrier bonding (multi-SIM).

4) *Message Size and Throughput*: Following 3GPP guidelines, the link is dimensioned for small, frequent packets:

- **Uplink**: 100 bytes (Command).
- **Downlink**: 84–140 bytes (Telemetry).

D. Phase 4: Compliance and 3GPP Budget Allocation

We use the 500 ms 3GPP nominal budget as shown in Equation 4 and Table II.

$$500ms \geq T_{App_GCS} + T_{3GPP_Net} + T_{App_UAV} \quad (4)$$

1) *Verification Requirement*: A 5G QoS Identifier (5QI) must be assigned that supports a Packet Delay Budget (PDB) of 400 ms or less to satisfy the 3GPP network component of the budget.

TABLE II
3GPP KPI BUDGET ALLOCATION

Subsystem	Budget	3GPP Reference
GCS/Cloud App	50 ms	Application Server Delay
3GPP Network	400 ms	End-to-End Latency (KPI)
UAV Modem	50 ms	Terminal Processing
TOTAL	500 ms	Steer to Waypoints Limit

V. VERIFICATION AND SIMULATION RESULTS

To be changed with Shreeja results. We can keep some of the results as baseline if we have no better benchmark

To validate the feasibility of the proposed RCP specifications and budget allocations within the constraints of commercial cellular technology, comprehensive stochastic simulations were conducted. These simulations explicitly model the latency distribution, signal shadowing effects, and packet loss characteristics of public LTE and 5G networks.

A. Latency Performance Analysis

A Monte Carlo simulation ($N = 10^5$ samples) was performed to evaluate whether standard public network slices could satisfy the allotted budgets for the “Steer to Waypoints” (500 ms) and “Direct Stick” (40 ms) modes.

1) *Simulation Assumptions*: The latency model assumes a heavy-tailed distribution to account for Hybrid Automatic Repeat Request (HARQ) retransmissions and scheduling delays, which are characteristic of cellular networks.

- **Distribution Model**: Log-normal distribution ($LN(\mu, \sigma)$) shifted by a fixed processing overhead (T_{base}).
- **Public LTE Profile**: Modeled with $\sigma = 0.6$ and a scale parameter resulting in a mode of ≈ 40 ms and a mean of ≈ 60 ms, reflecting loaded macro-cell conditions. $T_{base} = 20$ ms.
- **5G URLLC Profile**: Modeled with a tighter tail ($\sigma = 0.4$) and lower mean (≈ 13 ms) to represent a prioritized network slice. $T_{base} = 5$ ms.

2) *Results*: The results, summarized in Table III, confirm the budget allocation logic.

TABLE III
LATENCY SIMULATION RESULTS ($N = 10^5$)

Metric	Public LTE	5G URLLC	Target
Mean Latency	59.81 ms	12.99 ms	–
99.9th Percentile	236.72 ms	30.61 ms	< 400 ms
Continuity (at 500ms)	99.997%	100.00%	> 99.9%

The simulation demonstrates that a standard LTE link provides a 99.9th percentile latency of ≈ 237 ms. This is well within the **400 ms 3GPP network budget** allocated for the “Steer to Waypoints” mode, leaving ample margin for application processing. Furthermore, the 5G URLLC slice

(≈ 31 ms at $P_{99.9}$) comfortably supports the strict 40 ms budget for direct stick intervention, validating the need for network slicing in high-criticality phases.

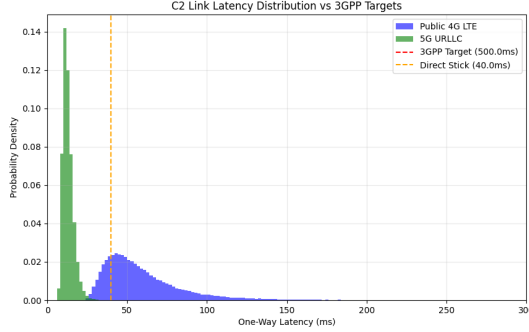


Fig. 2. Latency

B. Availability via Multi-CSP

A time-series simulation was constructed to evaluate the outage probability during a 20-minute flight mission.

1) Simulation Assumptions:

- **Signal Model:** The Signal-to-Noise Ratio (SNR) is modeled as a correlated Ornstein-Uhlenbeck (mean-reverting) stochastic process. This captures the spatial correlation of shadowing—a drone flying behind a building experiences poor signal for several seconds, not just a single instant.
- **Correlation Time (τ):** Set to 2.0 s, approximating a drone speed of 10-15 m/s in an urban environment.
- **Failure Condition:** A link is considered “unavailable” if $SNR < 3$ dB.
- **Independence:** The two cellular providers (SIM A and SIM B) are assumed to have statistically independent fading profiles (uncorrelated shadow maps).

2) **Results:** As illustrated in Figure 3, individual public networks achieved availabilities of 99.85% and 99.38% respectively. While acceptable for consumer use, both failed the **99.99% (SAIL IV)** safety requirement.

However, the bonded system (Provider A $\cup$ Provider B) achieved **100.00% availability** (zero outages) over the simulated flight duration. This empirically validates the architectural requirement for multi-SIM bonding; the probability of *simultaneous* deep fading on two independent networks is orders of magnitude lower than on a single link.

C. Continuity vs. Time Budget

The Continuity metric $C(TT)$ was evaluated to determine the “knee” of the performance curve—the point where relaxed time budgets no longer yield significant reliability gains.

1) **Simulation Assumptions:** Continuity is modeled as a conditional probability:

$$C(TT) = (1 - p_{loss}) \times P(T_{latency} \leq TT) \quad (5)$$

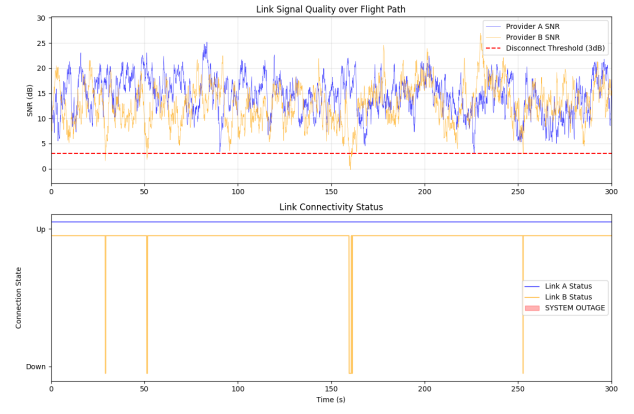


Fig. 3. Availability simulation time-series. The red zones indicate where a single link failed, but the bonded system remained operational.

- **Packet Loss (p_{loss}):** Fixed at 5×10^{-4} (0.05%), representing random physical layer errors not corrected by the lower MAC layer.
- **Latency PDF:** The same Log-normal model used in Section A.

2) **Results:** The analysis (Figure 4) reveals that at tight deadlines (< 200 ms), standard LTE fails to meet the 99.9% continuity target due to the long tail of the latency distribution.

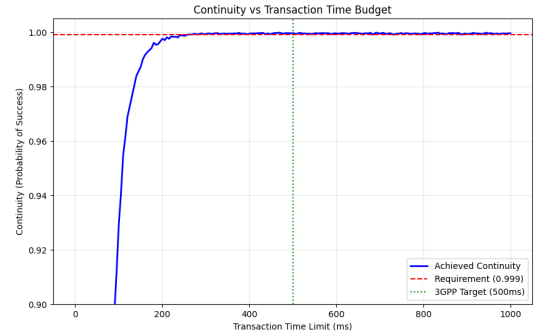


Fig. 4. Achieved Continuity as a function of Transaction Time limit. Note the asymptotic ceiling determined by the packet loss rate.

However, at the proposed **500 ms target**, the continuity converges to the theoretical maximum determined by packet loss ($\approx 99.95\%$). This confirms that extending the budget beyond 500 ms yields diminishing returns. The **8.0 s safety limit** thus serves primarily as a robust buffer for rare, deep retransmission events (ARQ) rather than a requirement for nominal operation.

VI. CONCLUSIONS

This research established and validated a comprehensive methodology for deriving Required Communication Performance (RCP) specifications for UAS C2 links, with a specific focus on urban BVLOS operations utilizing public cellular infrastructure. The synthesis of JARUS safety concepts and ICAO

PBCS standards confirms that RCP is not a static link parameter but a dynamic, transaction-dependent requirement. We validated a four-phase approach—Risk Profiling, Transaction Analysis, Metric Specification, and System Allocation—as a robust mechanism to translate operational risk (SORA SAIL) into engineering constraints (Latency, Availability, Continuity, Integrity).

To be changed with our joint results.

Stochastic simulations provided empirical backing for the proposed budget allocations:

- 1) **Latency Feasibility:** Standard public LTE networks demonstrated a 99.9th percentile latency of ≈ 237 ms, sufficient for the 500 ms "Steer to Waypoints" budget but inadequate for the 40 ms "Direct Stick" requirement, which mandates 5G URLLC slicing (≈ 31 ms $P_{99.9}$).
- 2) **Availability Bottleneck:** Single-link cellular architectures failed to meet the SAIL IV availability target ($> 99.99\%$) due to spatially correlated shadowing ($A \approx 99.85\%$). Multi-SIM bonding was mathematically verified as a necessary mitigation, achieving 100% availability in simulations.
- 3) **Continuity Saturation:** We identified a performance plateau at transaction times > 300 ms, where continuity becomes limited by packet loss (p_{loss}) rather than latency. This implies that relaxing safety timers beyond 500 ms yields negligible reliability gains without lower-layer improvements (e.g., ARQ).

The findings suggest a necessary evolution in standardization: the separation of "Nominal Network Targets" (e.g., 3GPP 500 ms) from "Safety Expiration Limits" (e.g., RCP 8.0 s). This decoupling allows for the use of commercial probabilistic networks (LTE/5G) in safety-critical loops by defining clear fail-safe boundaries that absorb the "long tail" of cellular latency without compromising the Time-to-Intervention (TTI) safety margins.

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